

secure and some may be malicious. MAM ensures that any new application installed is managed through proper channels. This way, only apps with the right permissions are allowed to be installed.

Not all applications are dangerous but those that are not required for business, such as games and social media apps, are denied installation.

Unified endpoint management (UEM): UEM is an evolution of MDM, managing both mobile and non-mobile devices. Through UEM, end users can use different types of devices. Applications can be used across different platforms.

Mobile protection measures

Mobile protection measures are like your device's personal bodyguards, ensuring its safety.

Remote wipe: Managed by MDM, remote wipe removes all the data from the device, usually during theft, necessitating proper data back-up.

Geolocation: This uses location tracking systems for various purposes, including during commute, cross-border alerts and finding lost phones.

Geofencing: This involves implementing restrictions on mobile features when the device is present in a particular location, with authentication and login allowed only within designated areas.

Screen lock: Mobiles are secured using a PIN, passcode, or biometrics, with auto-lock features after a configured time, and data erasure after multiple invalid entries.

Push notification services: This is an information pop-up service on the screen, receiving notifications even when the phone is idle or is using a different application.

Passwords, PINs and biometrics: This involves multi-authentication for mobile devices based on the apps used, managed through MDM, with password rotation, reset and complexity policies.

Context-aware authentication: This helps switching to multi-level authentication when abnormalities are

observed, such as access through different locations, different Wi-Fi connections, or paired Bluetooth devices.

Containerisation: This segments storage for business use to avoid data leaks and facilitate easy offboarding, while retaining personal data.

Full device encryption: This ensures data security even after its theft, with keys managed and rotated through MDM.

MicroSD hardware security module (HSM): This provides security services like encryption, key generation, rotation, digital signatures, and encrypted key storage.

SEAndroid (Security Enhancements for Android): This is built using SELinux (Security Enabled Linux) with centralised policy management. It ensures devices are patched with the latest security fixes.

Mobile deployment models

Enterprise mobile device deployment models fall into any of the following categories.

BYOD (Bring Your Own Device): Employees own the device, which can be challenging to secure, and containerisation through MDM can be used to manage corporate data.

COPE (Company Owned/ Personally Enabled): Company buys the device, which is for both professional and personal use. The organisation has full control on the device, including monitoring.

CYOD (Choose Your Own Device): Similar to COPE, these are corporate-owned devices where the user selects the mobile device. This can get tricky in terms of security for certain models.

COBO (Company Owned/ Business Only): Company buys devices and uses them only for official purposes with specific restrictions.

VMI (Virtual Mobile Infrastructure): Apps and data are separated from the mobile device. Data is stored securely in a centralised location to minimise risk.

Mobile security, along with other general security measures, plays a crucial role in ensuring the safety of enterprises and individuals in the rapidly evolving digital landscape. By employing robust mobile management and protection measures, businesses can safeguard sensitive data and mitigate potential threats effectively. 

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